Exercise 4:

1. SQL Joins

```
1. SELECT EmployeeID,
          firstname,
          Lastname,
          Department,
          Salary,
          ProjectID,
          ProjectName,
          Budget,
          Status

FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	firstname	Lastname	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	ongoing
3	Bob	Johnson	finance	75000	104	Financial Analysis	90000	ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	completed
6	Micheal	Clark	finance	80000	106	Risk Management	80000	pending

2. SELECT Empolee ID,
first name,
Last name,
Department,
Salary,
ProjectID,
Project name,
Budject,
Status
From Employees AS A
Left JOIN Projects AS B
On A.EmployeeID = B.EmployeeID;

Employee ID	firstname	Lastname	Department	Salary	Project ID	Project Name	Budged	Status
1	John	Doe	IT	70 000	101	AI Dev	100 000	Completed
1	Jonh	Doe	IT	70 000	103	cybersecurity Audit	75 000	Pending
2	Alice	Smith	HR	60 000	102	Employee Traing	~~78~~ 50 000	Ongoing
3	Bob	Johnson	finance	75 000	104	Financial Analysis	90 000	Ongoing
4	David	Brown	IT	72 000	NULL	Null	NULL	NULL
5	Emma	Wilson	Sales	65 000	105	Market Expansion	65 000	Completed
6	Micheal	Clark	finance	80 000	106	Risk Management	80 000	Pending

3. SELECT projectID,
ProjectName,
Budget,
Status,
EmployeeID,
firstname,
Lastname,
Department,
Salary

from Employees AS A
RIGHT JOIN projects AS B
ON A.EmployeeID = B.EmployeeID;

Project ID	Project Name	Budget	Status	Employee ID	firstname	Last Name	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70 000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60 000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70 000
104	financial Analysis	90000	Ongoing	3	Bob	Johnson	finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65 000
106	Risk Management	80000	Pending	6	Micheal	Clark	finance	80 000

4. SELECT EmployeeID,
firstname,
Lastname,
Department,
Salary,
ProjectID,
Projectname,
Budget,
Status

FROM Employees AS A
FULL OUTER JOIN projects AS B
On A.Employee ID = B. Employee ID

Emp ID	firstName	Last Name	Dept.	Salary	Project ID	Projectname	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	finance	75000	104	financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	sales	65000	105	Market Expansion	65000	completed
6	Micheal	Clark	finance	80000	106	Risk Management	80000	pending

2. UNIONS

5. SELECT City AS Location
FROM Employees
UNION
SELECT Status
FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

6. SELECT city AS location
FROM Employees
UNION ALL
SELECT Status
FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Ongoing
Completed
Pending

3. filtering Statements

7. SELECT Employee ID,
first name,
Last Name,
Department,
Salary
FROM Employees
WHERE Salary > 70000;

Employee ID	firstName	lastName	Department	Salary
3	Bob	Johnson	finance	75000
4	David	Brown	IT	72000
6	Micheal	Clark	finance	80000

8. SELECT firstName, LastName, EmployeeID, Salary, Department

FROM Employees

WHERE Department IN ('IT', 'finance');

employee ID	firstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	finance	75 000
4	David	Brown	IT	72 000
6	Micheal	clark	finance	80 000

9. SELECT ProjectID,
ProjectName,
Budget,
Status

from Projects

WHERE Status != 'Completed';

ProjectID	ProjectName	Budget	status
2	Employee Training	50000	Ongoing
1	Cybersecurity Audit	75000	Pending
3	financial Analysis	90000	Ongoing
6	Risk Management	80000	Pending

10. SELECT ProjectID,
ProjectName,
Budget,
Status

FROM Projects

WHERE Budget > 70 000

AND Status != 'Completed';

ProjectID	ProjectName	Budet	Status
1	Cyberssecurity Audit	~~80~~ 75000	Pending
3	Financial Analysis	90000	Ongoing
6	Risk Management	80000	Pending

11. SELECT EmployeeID,
firstname,
LastName,
Department,
Salary,
City
FROM Employees
WHERE ('New York', 'Toronto')
ORDER BY Salary DESC;

EmployeeID	firstname	LastName	Dept.	Salary	City
6	Micheal	Clark	finance	80000	New York
3	Bob	Johnson	finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. SELECT Employee ID,
firstname,
Last name,
Department,
Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	firstname	LastName	Department	Salary
6	Micheal	Clark	finance	80000
3	Bob	Johnson	finanance	75000
4	David	Brown	IT	72000

H. Aggregate functions with GROUP BY & HAVING
B. SELECT SUM (Salary) AS total_Salary,
Department
FROM Employees
GROUP BY Department
ORDER BY total_Salary DESC;

Department	total_Salary
finance	155 000
IT	142 000
Sales	65 000
HR	60 000

14. SELECT City,
AVG (Salary) AS Average_Salary
from Employees
Group By City
HAVING AVG(Salary) > 65000;

City	Average_Salary
New York	75 000
Toronto	75000

15. SELECT COUNT (EmployeeID) AS EmployeeCount,
Department
from Employees
Group BY Department
HAVING Count (EmployeeID) > 1;

Department	EmployeeCount
IT	2
finance	2

16. SELECT Count (ProjectID) AS ProjectCount,
Status
from Projects
Group BY Status
HAVING Count (ProjectID) >= 2;

Status	Project Count
Ongoing	2
Pending	2
Completed	2

17 SELECT EmployeeID, firstName, LastName,
sum (Budget) AS TotalProjectBudget
from Employees AS A
JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY EmployeeID, firstName, LastName
HAVING SUM(Budget) > 150 000;

EmployeeID	firstName	LastName	TotalProjectBudget
1	John	Doe	175 000